Supporting Information for

Integrated sensing and communication based on space-time-coding metasurfaces

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Supplementary Table 1

Supplementary Note 1: Details on Space-Coding Design

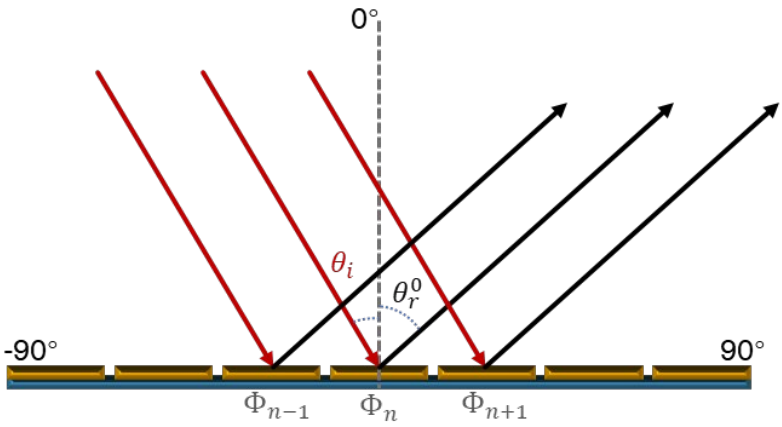
As illustrated in Supplementary Fig. 1, the space-time coding metasurface (STCM) is illuminated by a plane wave at an angle of incidence θ_i, with the direction of the desired beam being θ_r^0. The phase difference required between adjacent unit cells to achieve this anomalous reflection is expressed as:

$$\Delta\Phi = -k_c d(\sin\theta_r^0 + \sin\theta_i) \tag{S1}$$

where the symbols are defined in the main text. Therefore, the phase response of the nth-column of unit cells can be expressed as

$$\Phi_n = -(n-1)k_c d(\sin\theta_r^0 + \sin\theta_i) \tag{S2}$$

For a 2-bit programmable metasurface, the phase response of each unit cell can take four discrete values (-π/2, 0, π/2, and π), which can be represented digitally as "0", "1", "2", and "3". Consequently, the phase response Φ_n of each unit cell in a column needs to be quantized into these four discrete values and encoded into the corresponding digital code to create the space-coding (SC) sequence.



Supplementary Fig. 1 | Schematic of the STCM-enabled anomalous reflection.

Supplementary Note 2: Details on Time-Coding Sequences

We consider a 2-bit reflection-type STCM consisting of N columns of unit cells. The reflection coefficient of each unit cell of the nth-column is represented by a periodic function

$$\Gamma_{\mathrm{T}}^{n}(t) = \sum_{l=1}^{L} \Gamma_{\mathrm{T}}^{(n,l)} G_l(t) \tag{S3}$$

where L is the length of the time-coding sequence, and $G_l(t)$ denotes an impulse train

function with period T_0, which, in a period, can be expressed as

$$G_l(t)=\begin{cases}1, & (l-1)\tau \le t \le l\tau \\ 0, & \text{otherwise}\end{cases} \tag{S4}$$

The periodic reflection coefficient $\Gamma_{\mathrm{T}}^{n}(t)$ can then be expanded into a Fourier series, with the generic coefficient expressed as

$$a_v^n=\sum_{l=1}^{L}\frac{\Gamma_{\mathrm{T}}^{(n,l)}}{L}\mathrm{sinc}\left(\frac{\pi v}{L}\right)\exp(-j\pi v(2l-1)/L) \tag{S5}$$

Supplementary Note 3: Details of the STC Matrix Design for Optimizing the Harmonics and Suppressing the Fundamental-Frequency Wave

The optimization of the STC matrix targets three key objectives for shaping the spatial- spectral characteristics of the harmonics. Firstly, it aims to enhance the orthogonality of the input vectors corresponding to different angles of incidence, thus ensuring a robust and generalized mapping from $\mathbf{s}(\theta_i)$ to θ_i. The optimized STC matrix, as displayed in Supplementary Fig. 2a, successfully achieves this objective, with the inner products between the normalized amplitude vectors $\mathbf{s}(\theta_i)$ for various angles shown in Supplementary Fig. 2b, hence confirming their orthogonality.

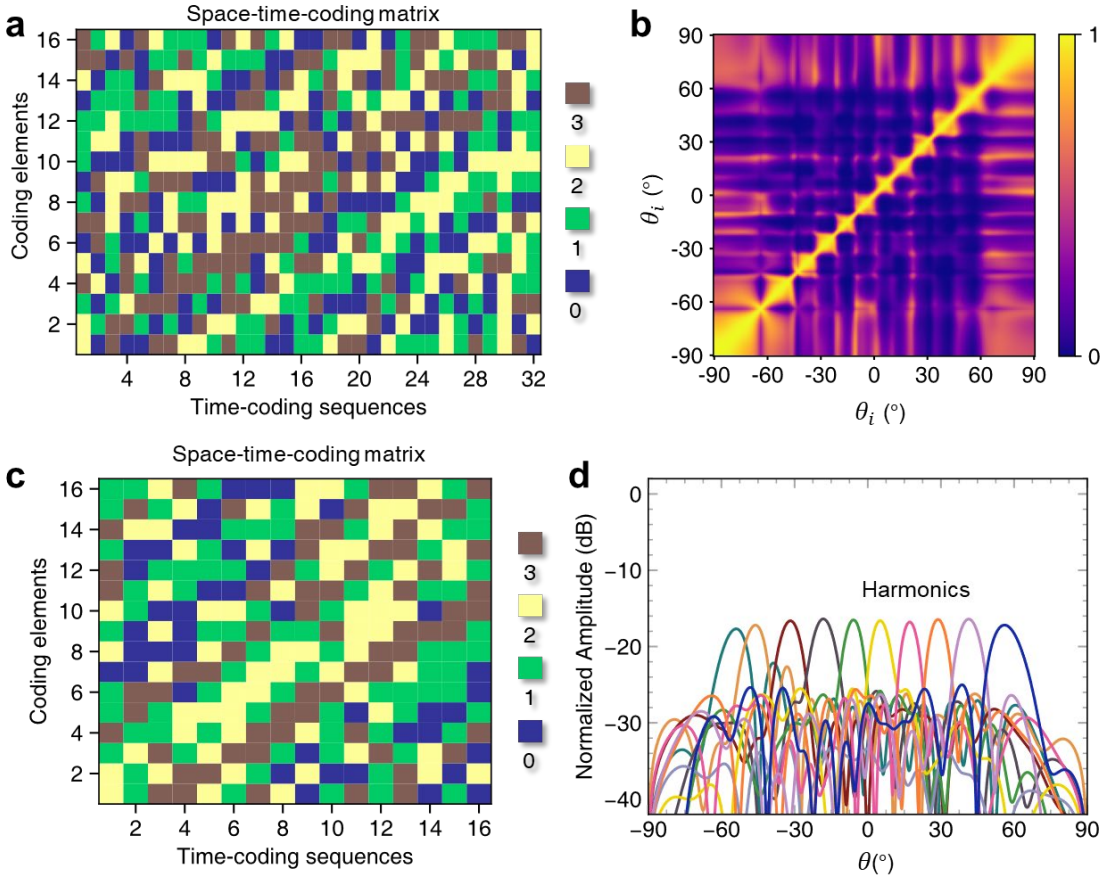
Secondly, the optimization scheme seeks to minimize the fundamental-frequency component in the reflected wave from the spatio-temporal-modulated region. This is crucial to reduce the interference with the beam steering capabilities of the STCM. Thirdly, it is essential to ensure that the receiving antenna captures a large amount of power from the harmonics at different angles of incidence. An insufficient received power could lead to a low signal-to-noise ratio, which would significantly reduce the accuracy of the estimated angle.

We exploit the binary particle swarm optimization (BPSO) algorithm to optimize the STC matrix, based on the following fitness function:

$$fitness=w_1\cdot\left(\sum_{v=\pm1,\pm2,\ldots\pm5}\left|\theta_s^v-\theta_d^v\right|\right)+w_2\cdot\left[\sum_{v=\pm1,\pm2,\ldots\pm5}(SLL_{\max}^v-SLL_d)+(UFL_{\max}^0-UFL_d)\right] \tag{S6}$$

Here, for any vth harmonic frequency, θ_s^v and θ_d^v denote the simulated and desired direction, respectively, of the main beams, $SLL_{\max}^v$ and SLL_d represent the maximum values of the simulated and desired sidelobe levels, respectively, and $UFL_{\max}^0$ and UFL_d represent the

simulated and desired, respectively, maximum levels of signal at the fundamental frequency. Moreover, w_1 and w_2 are weighting factors. The optimization process is carried out iteratively until a satisfactory STC matrix is determined. Additionally, the length L of time-coding sequences for optimization is not fixed. An example of an STC matrix with L=16 and the corresponding far-field scattering patterns are illustrated in Supplementary Fig. 2c and 2d.



Supplementary Fig. 2 | STC matrix design for optimizing the harmonics and suppressing the fundamental-frequency wave. a, Optimized STC matrix with L=32. **b**, Inner product between normalized-amplitude vectors corresponding to different angles of incidence. **c**, Optimized STC matrix with L=16. **d**, Corresponding far-field scattering patterns.

Supplementary Note 4: Details on Design of STC Matrix for Simultaneously Optimizing the Fundamental-Frequency Wave and Harmonics

Here, we introduce the design of STC matrices that allow the optimization of the propagation of the fundamental-frequency wave while generating multiple harmonics with distinct spatial distributions. Let us consider a 2-bit programmable metasurface consisting of N-column unit cells. The reflection coefficient of the nth-column in one period T_0 is expressed as follows

$$\Gamma_{\mathrm{T}}^{n}(t)=\begin{cases}\exp(j\pi), & t_s^n \le t \le t_s^n+\tau^n \\ 1, & \text{otherwise}\end{cases} \tag{S7}$$

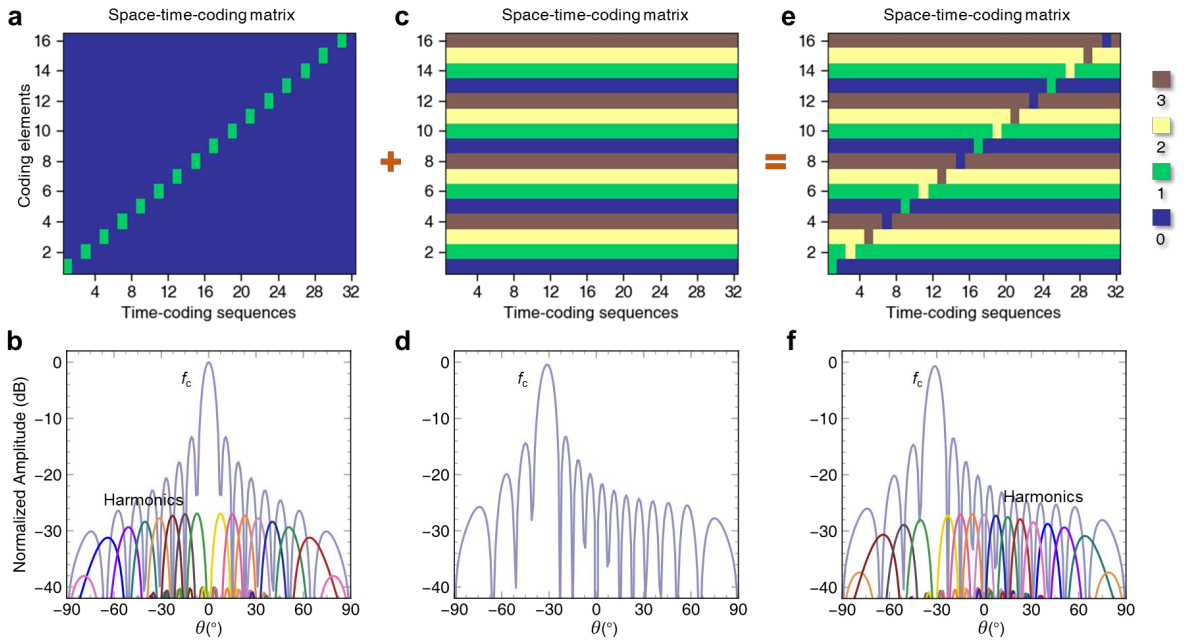
where t_s^n and τ^n represent the time shift and duty cycle, respectively, of the time-coding sequence for the *n*th-column. For example, with *N*=16, $t_s^n=(n-1)T_0/16$, $\tau^n=\tau_0=T_0/32$, the corresponding STC matrix is shown in Supplementary Fig. 3a. This matrix illustrates that all the *N*-column unit cells share the same duty cycle τ_0, with $\Gamma_{\mathrm{T}}^{n+1}(t)=\Gamma_{\mathrm{T}}^{n}(t-T_0/16)$. The periodic reflection coefficient $\Gamma_{\mathrm{T}}^{n}(t)$ can be expanded into a Fourier series as:

$$a_v^n=\frac{\tau_0}{T_0}\mathrm{sinc}\left(\frac{\pi v\tau_0}{T_0}\right)\exp\left(\frac{-j2\pi vnt_s}{T_0}\right) \tag{S8}$$

where $t_s=T_0/16$. Considering $a_v^{n+1}/a_v^n=\exp(-j2\pi vt_s/T_0)$, it can be observed that a_v^{n+1} and a_v^n have the same amplitude, but due to the time shift t_s, their phase difference is $2\pi vt_s/T_0$. Consequently, when the STCM is illuminated by a plane wave with an incidence angle $\theta_i=0^\circ$, the beam-pointing direction of the *v*th-order harmonic in the reflected wave can be expressed as:

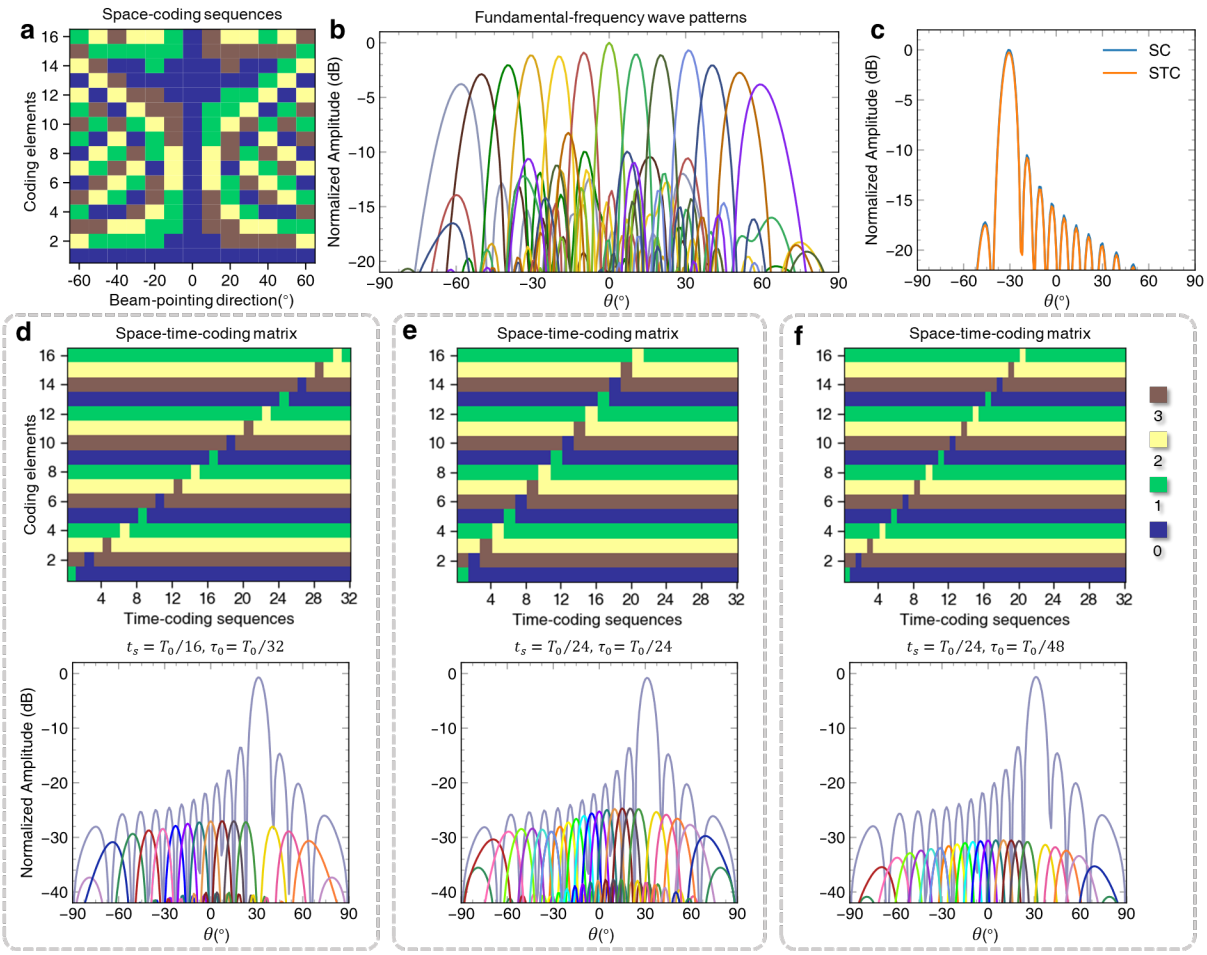
$$\theta_r^v=\arcsin\left(\frac{2\pi vt_s}{k_c dT_0}\right) \tag{S9}$$

Thus, different harmonics exhibit distinct beam-pointing directions, as demonstrated by the far-field scattering pattern shown in Supplementary Fig. 3b. To further control the beam-pointing direction of the fundamental-frequency wave θ_r^0, we calculate the phase response of the *n*th-column unit cells using Eq. (S2), given the desired θ_r^0 and the incidence angle θ_i. For instance, with $\theta_r^0=-60^\circ$, $\theta_i=0^\circ$, the corresponding SC pattern is represented by a time-invariant STC matrix, as illustrated in Supplementary Fig. 3c. The corresponding far-field scattering pattern of the fundamental-frequency wave is shown in Supplementary Fig. 3d. By combining (adding) the two STC matrices as illustrated in Supplementary Figs. 3a and 3c, we obtain a new STC matrix, shown in Supplementary Fig. 3e. The far-field scattering patterns of both the fundamental-frequency and harmonic waves are presented in Supplementary Fig. 3f. As can be observed, this new STC matrix effectively directs the fundamental-frequency wave to the desired angle $\theta_r^0=-60^\circ$ while generating multiple harmonics with distinct beam-pointing directions.



Supplementary Fig. 3 | STC matrix design for simultaneously manipulating the fundamental-frequency wave and generating multiple harmonics. a, STC matrix for *N*=16, $t_s^n = (n-1)\,T_0/16$, $\tau^n = \tau_0 = T_0/32$. **b,** Corresponding far-field scattering patterns. **c**, Time-invariant STC matrix corresponding to a SC pattern with $\theta_r^0 = -30^\circ$. **d**, Corresponding far-field scattering patterns. **e**, New STC matrix obtained by combining (adding) the two STC matrices above. **f**, Corresponding far-field scattering patterns.

Assuming that the desired angle of reflection θ_r^0 varies from -60° to +60° in 10° increments and $\theta_i = 0^\circ$, the corresponding SC sequences are illustrated in Supplementary Fig. 4a. Based on these sequences, and assuming $t_s = T_0/16$ and $\tau_0 = T_0/32$, a series of STC matrices can be generated, with the resulting far-field scattering patterns of the fundamental-frequency wave shown in Supplementary Fig. 4b. When comparing the far-field scattering pattern produced by SC modulation and STC modulation, it is evident that a portion of the incoming wave's energy is converted into harmonics, while the gain of the fundamental-frequency wave decreases only slightly, as shown in Supplementary Fig. 4c. Moreover, by optimizing the duty cycle τ_0 and the time shift t_s, the beam-pointing directions and power of the harmonics can be finely tuned. For instance, assuming $\theta_r^0 = 30^\circ$, Supplementary Fig. 4d-f present three STC matrices using the parameters $t_s = T_0/16$, $\tau_0 = T_0/32$; $t_s = T_0/24$, $\tau_0 = T_0/24$; and $t_s = T_0/24$, $\tau_0 = T_0/48$ and corresponding far-field scattering patterns.

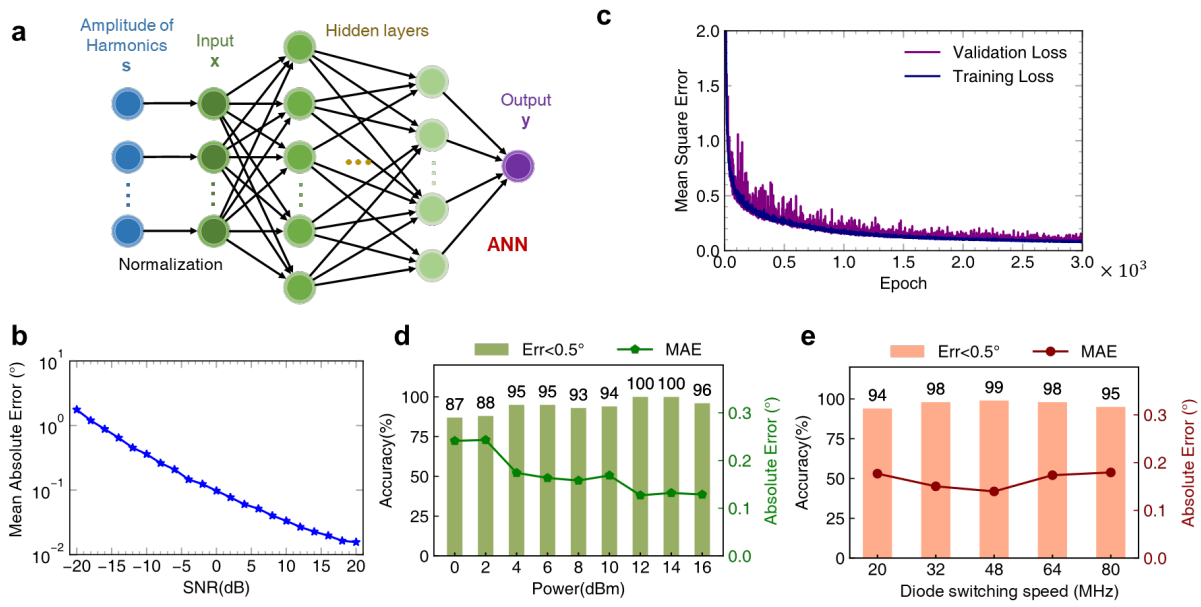


Supplementary Fig. 4 | STC matrices corresponding to different values of θ_r^0, τ_0, and t_s. a, SC sequences as a function of θ_r^0. **b**, Far-field scattering patterns of the fundamental-frequency wave corresponding to the generated STC matrices. **c**, Far-field scattering pattern of the fundamental-frequency wave obtained via SC and STC modulation. **d-f**, Three representative STC matrices corresponding to the parameters (left to right): $t_s = T_0/16$, $\tau_0 = T_0/32$; $t_s = T_0/24$, $\tau_0 = T_0/24$; and $t_s = T_0/24$, $\tau_0 = T_0/48$ and corresponding far-field scattering patterns.

Supplementary Note 5: Details on ANN Model

As illustrated in Supplementary Fig. 5a, we set up a 6-layer fully connected ANN architecture comprising an input layer with a dimension of 10, corresponding to the harmonics amplitude vector $\mathbf{x}(\theta_i)$ (from the -5th to the +5th harmonic orders), four hidden layers, and a 1-dimensional output layer corresponding to the estimated angle of incidence. The signal- to-noise ratio (SNR) is the primary factor determining the upper limit of DOA estimation accuracy. We evaluated DOA estimation performance across varying SNR levels, observing that the mean absolute error decreases gradually as SNR improves, as seen in Supplementary Fig. 5b.

To gather training data, the far-field scattering patterns of the STC matrix shown in Fig. 2c are measured across different values of the transmit powers and diode switching speeds. A total of 24 datasets are collected with transmit powers ranging from -5 dBm to 20 dBm, with a step of 5 dBm, and diode switching speeds of 32, 40, 64, and 80 MHz. From these datasets, 10% of the data is utilized as the validation set, another 10% of the data is utilized as the test set, and the remaining 80% of the data is utilized for training. Supplementary Figure 5c shows the mean square errors (MSEs) for the training and validation sets during the parameter optimization process. Convergence of the network parameters is observed after 3,000 epochs, with similar trends of the curves indicating no over-fitting. The performance of the ANN architecture for DOA estimation based on the test set, with varying transmit powers and PIN diode switching speeds, is presented in Supplementary Figs. 5d and 5e in terms of mean absolute error (MAE) and percentage of absolute error <0.5°. The results show a progressive decrease of the MAE from 0.24° to 0.13° as the transmit power increases from 0dBm to 16dBm, and an increase of the percentage of the absolute error <0.5°. However, there is no apparent trend based on the same data sets as a function of the switching speed of the diode.



Supplementary Fig. 5 | Architecture and performance of the ANN architecture. a, ANN architecture. **b**, Mean absolute error of estimation as a function of SNR. **c**, MSE curves for training and validation sets during the parameter optimization of the ANN model. **d**,**e**, Estimated performance of the ANN architecture based on test sets with varying transmit powers and PIN diode switching speeds, respectively.

Supplementary Note 6: Details on DOA Estimation Based on Sensing Matrix and Correlation Coefficient

By utilizing the proposed full-aperture coding strategy for the STCM, we can control the propagation of the fundamental-frequency wave while simultaneously generating multiple harmonics that are evenly distributed in space. We can also analyze the spatial-spectral characteristics of the harmonics for DOA estimation based on the sensing matrix and the correlation coefficient. For instance, let us consider the STC matrix $\mathbf{W}$ with $t_s = T_0/24$ and $\tau_0 = T_0/48$, as shown in Supplementary Fig. 6a. When an incoming wave with an angle of incidence θ_i illuminates the STCM, the amplitudes of the harmonics with orders v=±1, ±2…±M along the normal direction in the reflected wave are grouped in a vector $\mathbf{s}(\mathbf{W}, \theta_i)$ of dimension 2M. Supplementary Figure 6c presents the normalized amplitude distributions of the harmonics for $\theta_i = -50^\circ$ and $\theta_i = 20^\circ$. We can define a unit vector $\mathbf{y}(\mathbf{W}, \theta_i) = \mathbf{s}(\mathbf{W}, \theta_i) / \|\mathbf{s}(\mathbf{W}, \theta_i)\|$, and select K angles θ_i within the range [-90°, 90°], so that the sensing matrix $\boldsymbol{H}$ can be expressed as:

$$\mathbf{H}(\mathbf{W}) = \left[\mathbf{y}(\mathbf{W}, \theta_i^1), \mathbf{y}(\mathbf{W}, \theta_i^2), \cdots, \mathbf{y}(\mathbf{W}, \theta_i^K) \right]^{\mathrm{T}} \tag{S10}$$

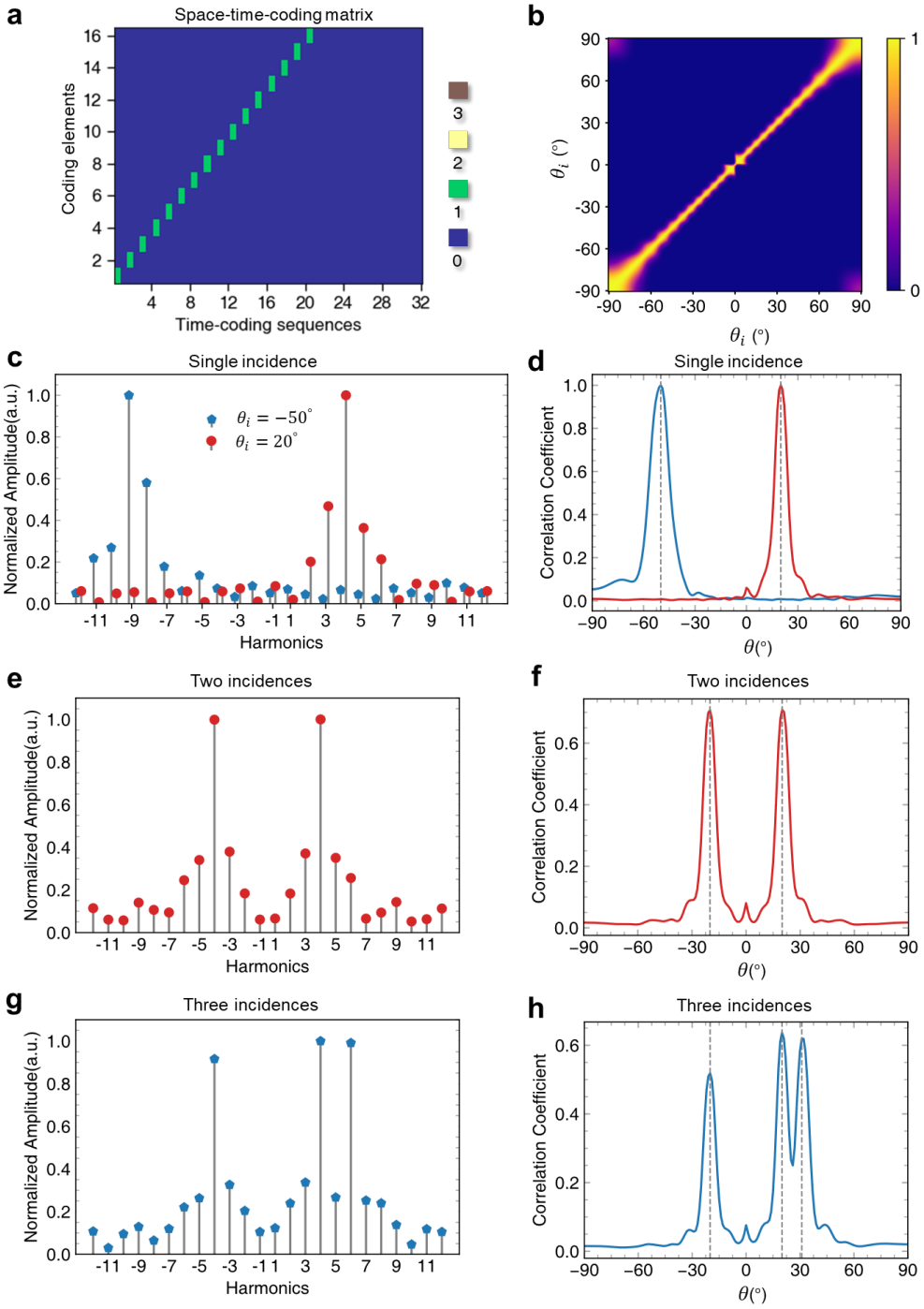
The obtained vectors in (S10) exhibit high orthogonality to one another, as demonstrated by calculating their inner products and shown in Supplementary Fig. 6b. We define the correlation coefficient as:

$$\mathbf{z}(\mathbf{W}, \theta_i) = \mathbf{H}(\mathbf{W}) \cdot \mathbf{y}(\mathbf{W}, \theta_i) \tag{S11}$$

The position of the peak in $\mathbf{z}(\mathbf{W}, \theta_i)$ corresponds to the angle of incidence θ_i. By setting $\boldsymbol{\theta} = [-90°, -89° \ldots, 90°]$, the correlation coefficients for $\theta_i = -50^\circ$ and $\theta_i = 20^\circ$ are shown in Supplementary Fig. 6d, where the peak positions clearly indicate the angle of incidence θ_i.

Moreover, the proposed DOA estimation method is also effective for multiple angles of incidence. For example, with $\theta_{i1} = -20^\circ$ and $\theta_{i2} = 20^\circ$, we calculate the amplitudes of the harmonic in the normal direction when two waves from two given directions illuminate the STCM simultaneously. Supplementary Figures. 6e,f display the corresponding distribution of the normalized amplitude of the harmonics and the correlation coefficients. As can be observed, the positions of the two peaks in the correlation coefficient correspond to the two angles of incidence. For three angles of incidence, the distribution of the normalized amplitude of the harmonics and the correlation coefficient for $\theta_{i1} = -20^\circ$, $\theta_{i2} = 20^\circ$, and $\theta_{i3} = 31^\circ$ are shown in Supplementary Figs. 6g,h. Once again, the three peaks in the correlation coefficient

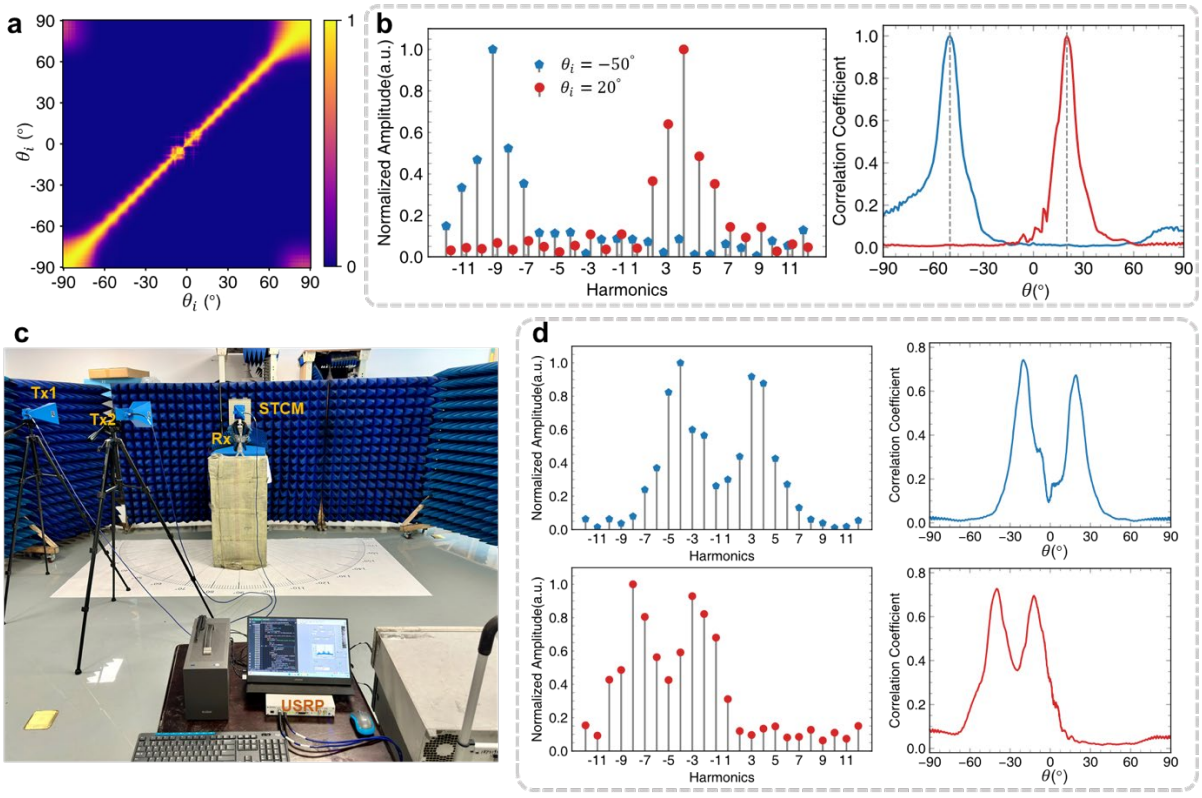
correspond to the three angles of incidence. As the number of incident beams increases, the accuracy may degrade, especially when the beams arrive from similar directions. To enhance the sensing accuracy for multiple beams, we can utilize additional harmonics with different spatial distributions and a larger metasurface to improve the angular resolution and mitigate the interference between closely spaced beams.



Supplementary Fig. 6 | DOA estimation method based on sensing matrix and correlation coefficient. a, Example of STC matrix with N=16, $t_s^n = (n-1)\,T_0/24$, and $\tau^n = \tau_0 = T_0/48$. **b**, Corresponding distribution of the normalized amplitude of the harmonics for $\theta_i = -50^\circ$ and $\theta_i = 20^\circ$. **c**, Inner products

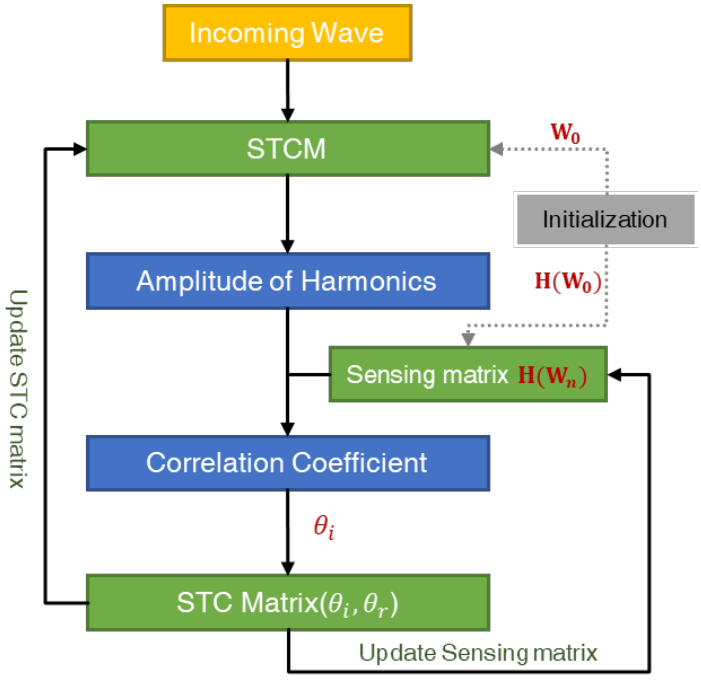
between the unit vectors. **d**, Correlation coefficients for $\theta_i = -50^\circ$ and $\theta_i = 20^\circ$. **e, f**, Distribution of normalized amplitude of harmonics and correlation coefficient, respectively, for two impinging waves ($\theta_{i1} = -20^\circ$ and $\theta_{i2} = 20^\circ$). **g, h**, Distribution of normalized amplitude of harmonics and correlation coefficient, respectively, for three impinging waves ($\theta_{i1} = -20^\circ$, $\theta_{i2} = 20^\circ$, and $\theta_{i3} = 31^\circ$).

To assess the effectiveness of the proposed DOA estimation method, we have tested the STCM in an anechoic chamber utilizing the STC matrix shown in Supplementary Fig. 6a. Specifically, we have measured the unit vectors of the amplitudes of the harmonics at different angles of incidence when a single source was present. These measured unit vectors also exhibit high orthogonality, as demonstrated by the computation of their inner products as shown in Supplementary Fig. 7a. The distribution of the normalized amplitudes of the harmonics and the correlation coefficients for $\theta_i = -50^\circ$ and $\theta_i = 20^\circ$ are displayed in Supplementary Fig.7b. These measured results are in very good agreement with the simulation results. We also conducted tests and measurements in an indoor setting in the presence of two sources, as shown in Supplementary Fig. 7c. The distribution of the normalized amplitudes of the harmonics and the correlation coefficients for two impinging waves ($\theta_{i1} = -20^\circ$, $\theta_{i2} = 20^\circ$; and $\theta_{i1} = -40^\circ$, $\theta_{i2} = -10^\circ$) are presented in Supplementary Fig. 7d. The peak positions in the correlation coefficients correspond to the two angles of incidence.



Supplementary Fig. 7 | Experimental setup and measurement results. a, Inner products between the measured unit vectors. **b**, Corresponding distribution of the normalized amplitudes of the harmonics and correlation coefficients for different values of θ_i. **c.** Indoor experimental setup for two angles of incidence. **d.** Corresponding distribution of the normalized amplitudes of the harmonics and correlation coefficients for two scenarios for three angles of incidence ($\theta_{i1} = -20^\circ$, $\theta_{i2} = 20^\circ$; and $\theta_{i1} = -40^\circ$, $\theta_{i2} = -10^\circ$, respectively).

Finally, as far as the full-aperture scheme is concerned, the STC matrix to control the propagation of the fundamental-frequency wave needs to be modified. Specifically, the proposed approach is based on the fact that different STC matrices correspond to distinct sensing matrices, as illustrated in Supplementary Fig. 8. The process begins with the STCM being configured using an initial STC matrix $\mathbf{W}_0$, and the corresponding sensing matrix is set to $\mathbf{H}(\mathbf{W}_0)$. The direction θ_i of the incoming wave is then determined by calculating the correlation coefficient. Based on θ_i and the desired direction of reflection θ_r, the STCM is updated with the STC matrix $\mathbf{W}_n(\theta_i, \theta_r)$ to steer the fundamental-frequency wave towards the target direction θ_r. Simultaneously, the sensing matrix is updated with $\mathbf{H}(\mathbf{W}_n)$. As the angle of incidence is continuously estimated, any changes in θ_i prompt the metasurface's STC matrix to be updated according the new θ_i and desired direction θ_r, thus updating the sensing matrix as well. This approach ensures that the incoming wave is consistently reflected towards the desired direction θ_r, as the angle of incidence changes.

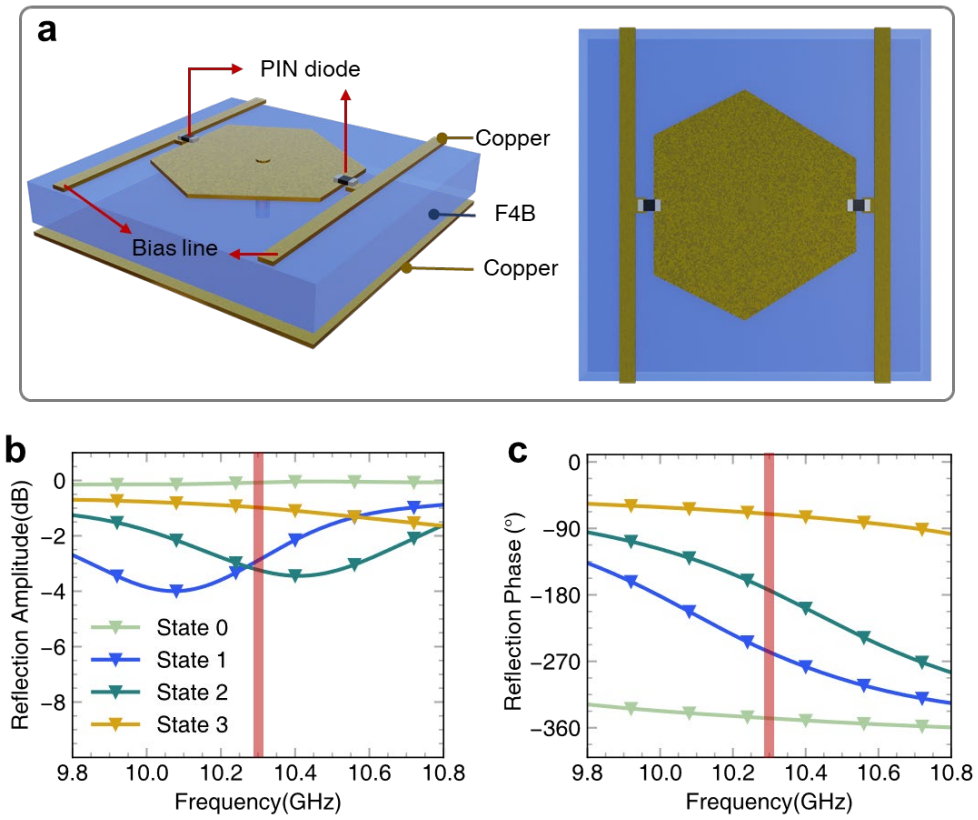


Supplementary Fig. 8 | Flowchart of the process for realizing self-adaptive beam-steering of the ISAC

system with the full-aperture scheme.

Supplementary Note 7: Details on the 2-Bit STCM and its Power Consumption

The prototype used for validating the proposed scheme is a 2-bit reflection-type STCM comprising 16×16 unit cells. The size of each unit cell is 14 mm×14 mm and is composed of a hexagonal metal patch and two biasing lines printed on a grounded F4B substrate (with relative permittivity $\varepsilon_r = 2.65$ and loss-tangent $\tan\delta = 0.001$), as shown in Supplementary Fig. 9a. Two PIN diodes are integrated to connect the hexagonal patch to the biasing lines. Supplementary Figures 9b and 9c show the measured amplitude and phase responses, respectively, of the unit cell in states "0", "1", "2", and "3".



Supplementary Fig. 9 | 2-bit reflection-type digital coding metasurface. a, Schematic of the unit cell. **b**,**c**, Measured amplitude and phase responses, respectively, of the unit cell in states "0", "1", "2", and "3".

When the forward current of PIN diodes reaches 1 mA, the forward voltage is 1.2 V, enabling the diodes to operate in the “ON” state, with a minimum power consumption of 1.2 mW per diode. When all PIN diodes in the metasurface are “ON”, the total power consumption of the programmable metasurface is approximately 0.61 W.

In our experiments, a general-purpose FPGA hardware board was used to control the coding states of unit cells, supplying a driving current of more than 1 mA to each diode. The combined power consumption of the metasurface and FPGA measured across various coding states is shown in Supplementary Table 1, with an average power consumption of approximately 4.34 W. If all PIN diodes are in the "ON" state, the total power consumption reaches about 7.09 W. Moving forward, a customized control module could be developed to further reduce power consumption.

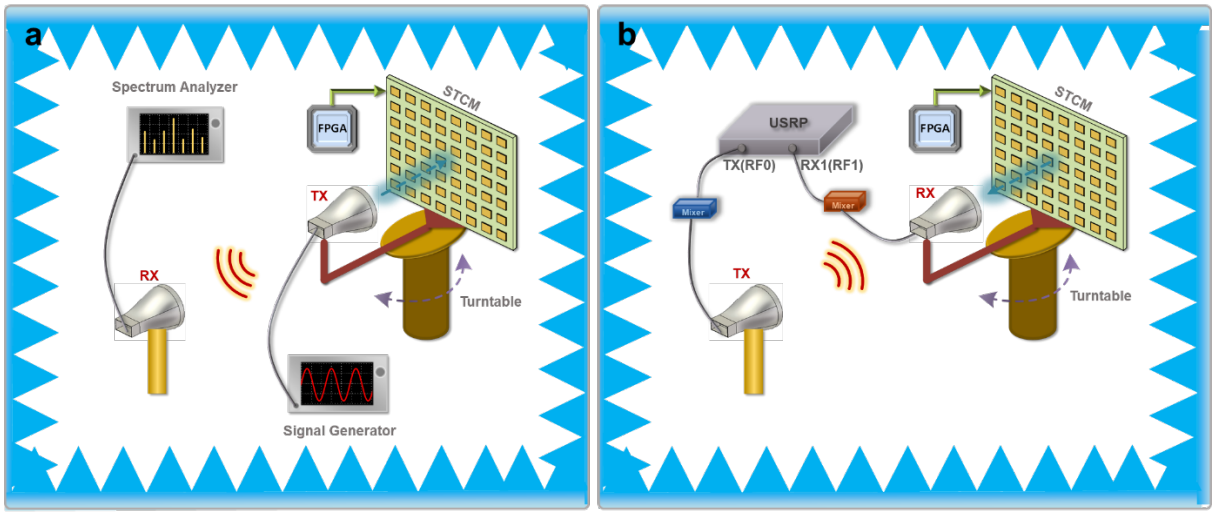
Supplementary Table 1. Total power consumption across various coding states

Coding State	Current (A)	Voltage (V)	Power (W)
0	0.07	5.13	0.36
1	0.98	5.06	4.96
2	0.98	5.06	4.96
3	1.41	5.03	7.09

Supplementary Note 8: Details on Experimental Setup in Anechoic Chamber

The experimental setup for measuring the far-field scattering patterns is sketched in Supplementary Fig. 10a. It features a linearly polarized horn antenna that is positioned at 0.85m facing normally to the metasurface. This antenna, connected to a signal generator (Keysight N9010A), provides a monochromatic plane wave at 10.3 GHz. Another linearly polarized horn antenna, operating from 8 to 12 GHz, is positioned 8.5 m away from the center of the turntable. The FPGA hardware board (ALTERA Cyclone IV) dynamically controls the coding pattern of STCM, which is mounted on a turntable.

Supplementary Fig. 10b illustrates the case study when a monochromatic plane wave is replaced by a USRP-modulated signal, as the incoming wave. The output port TX(RF0) of USRP is connected to the antenna via a mixer, while another antenna is connected to the input port RX1(RF1) of USRP via another mixer. In this way, the modulated signal is up-converted to 10.3 GHz and transmitted. After the STCM modulates the incoming wave, the generated harmonics are received by the antenna and then down-converted to the input port of the USRP for analyzing the signal spectrum.

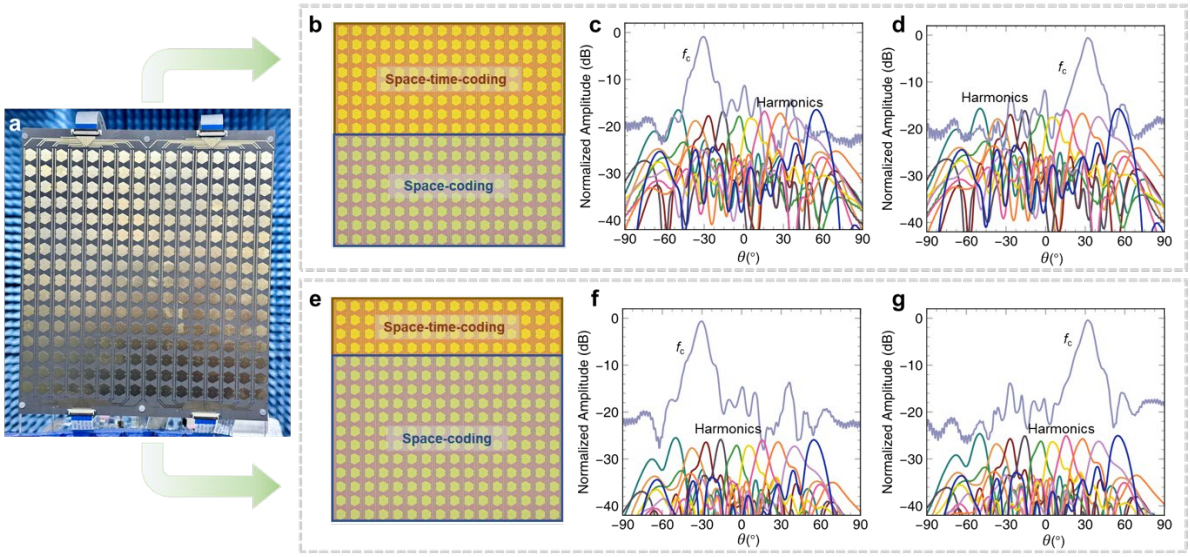


Supplementary Fig. 10 | Schematics of the experimental setups. a, Measurement of the far-field scattering patterns. **b**, Measurement and analysis of the USRP-modulated signal.

Supplementary Note 9: Details on STCMs with Different Partition Ratios

In addition to the STCM prototype that consists of two regions of equal area, we have developed another prototype featuring a finer partitioning, which is achieved through the optimization of the direct-current (DC) voltage bias network, as illustrated in Supplementary Fig. 11a. We tested this prototype with various partition ratios in an anechoic chamber. The STCM was initially divided into two equal-area regions, ensuring balanced power allocation for both wireless communication and sensing, as shown in Supplementary Fig. 11b. Supplementary Figs. 11c,d show the far-field scattering patterns at the fundamental and harmonic frequencies corresponding to different phase patterns applied to the SC region. The dynamic reconfigurability of the STCM allows for flexible adjustments of the areas and positions of the two regions, hence allowing the optimization of the power distribution for communication and sensing as needed. For instance, Supplementary Figs. 11f,g show the resulting far-field scattering patterns when one-quarter of the STCM is configured with an STC matrix and the remaining region with an SC matrix (see Supplementary Figs. 11e). As can be observed, the amplitudes of the harmonics are reduced.

Overall, the adjustable partitioning scheme for ISAC utilizes the same aperture and control module as the STCM, resulting in lower costs, reduced deployment space, and enhanced hardware integration.



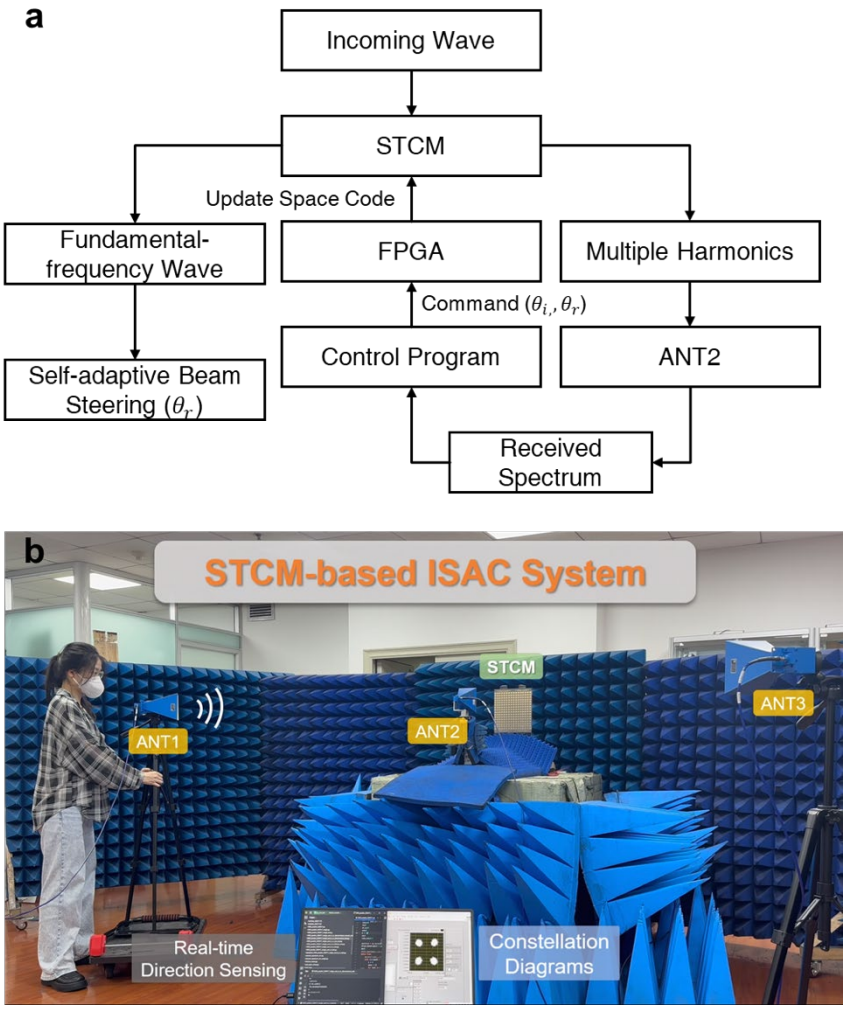
Supplementary Fig. 11 | STCM using different partitioning. a, STCM prototype with finer partition via optimization of the DC voltage bias network. **b**, Schematic of the STCM divided into two equal-area regions. **c**,**d**, Corresponding far-field scattering patterns at the fundamental and harmonic frequencies. **e**,**f**,**g** As in panels b,c,d, but for a partition corresponding to 1/4 STC and 3/4 SC.

Supplementary Note 10: Detailed Process for Realizing Self-Adaptive Operation

The STCM modulates the incoming wave, converting part of it into multiple harmonics. The receiving antenna, ANT2, is positioned perpendicular to the STCM and connected to the USRP's receiving channel RX1 (RF1), which analyzes the spectral distribution of the received harmonics. We developed a control program in Python that handles several key tasks: calculating the normalized harmonic vectors, running the ANN model for DOA estimation, generating control commands, and transmitting these commands to the FPGA that controls the STCM's coding pattern. The commands sent to the FPGA are based on the angle of incidence and the desired direction of beam-steering. When the direction of the incoming wave changes, the control program automatically sends new commands to the FPGA, which then adjusts the STCM's coding accordingly to ensure that the fundamental-frequency wave is appropriately steered toward the desired direction. The flowchart illustrating this process is shown in Supplementary Fig. 12a. Once initiated, the entire process operates automatically and continuously, without requiring human intervention.

The self-adaptive process is also demonstrated in Supplementary Movie 1, as demonstrated in Supplementary Fig. 12b. In the video, ANT1 is continuously moved to change the angle of incidence. The system detects the direction of the incoming wave in real-time, sending appropriate commands to FPGA for adjusting the coding pattern, so that the communication

link is maintained to the specified quality of service regardless of the directions of the incoming waves.



Supplementary Fig. 12 | a, Flowchart of the process for realizing self-adaptive beam-steering. **b**, Representative frame from Supplementary Movie 1.

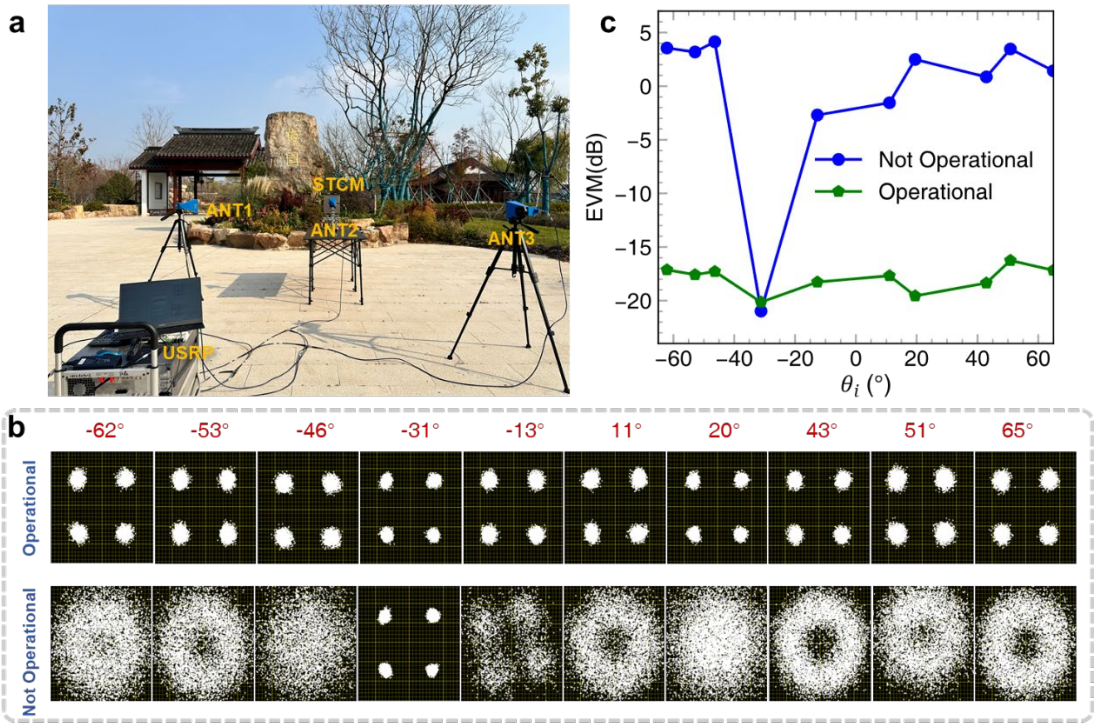
Supplementary Note 11: Validation in Real-World Scenarios

In previous experiments, system validation was conducted in an indoor environment surrounded by absorbing materials. However, the proposed STCM-based ISAC system is designed to operate reliably even in non-ideal conditions. To further assess its effectiveness, additional experiments were performed in two challenging scenarios: a realistic outdoor environment (illustrated in Supplementary Fig. 13) and an indoor environment without absorbing materials (shown in Supplementary Fig. 14).

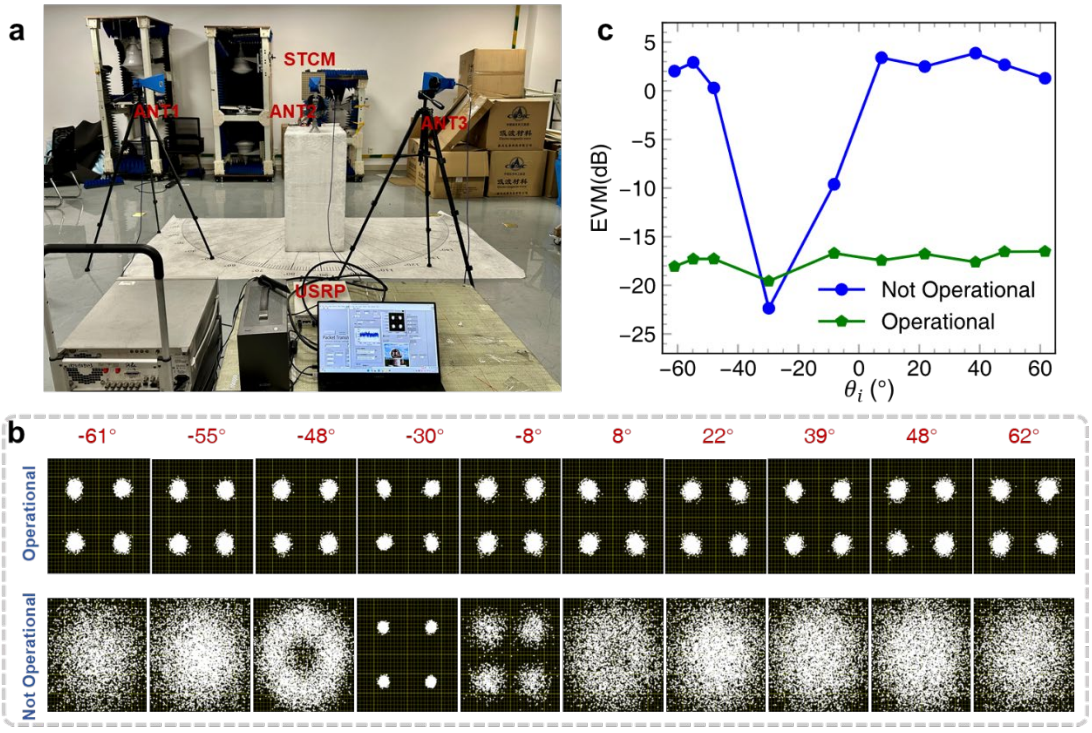
In both cases, the experimental setups replicated the original configuration depicted in Fig. 6. Antenna ANT1 was placed at 10 randomly selected measurement points to transmit the

modulated signal. The results revealed that when the STCM was inactive, the error vector magnitude (EVM) remained low only at angles of incidence that were specularly symmetrical with respect to antenna ANT3. In contrast, when the STCM was active, the demodulated constellation diagrams consistently exhibited high quality, and the EVM was kept at low levels across all measurement points.

These findings confirm that the STCM-based ISAC system maintains reliable performance in real-world and non-ideal environments.



Supplementary Fig. 13 | Validation of the STCM-based ISAC system in a realistic outdoor environment. a, Experimental setup. **b**, Demodulated constellation diagrams for 10 positions of antenna ANT1 when the STCM is not operational and operational. **c**, Measured EVM for different angles of incidence when the STCM is not operational and operational.

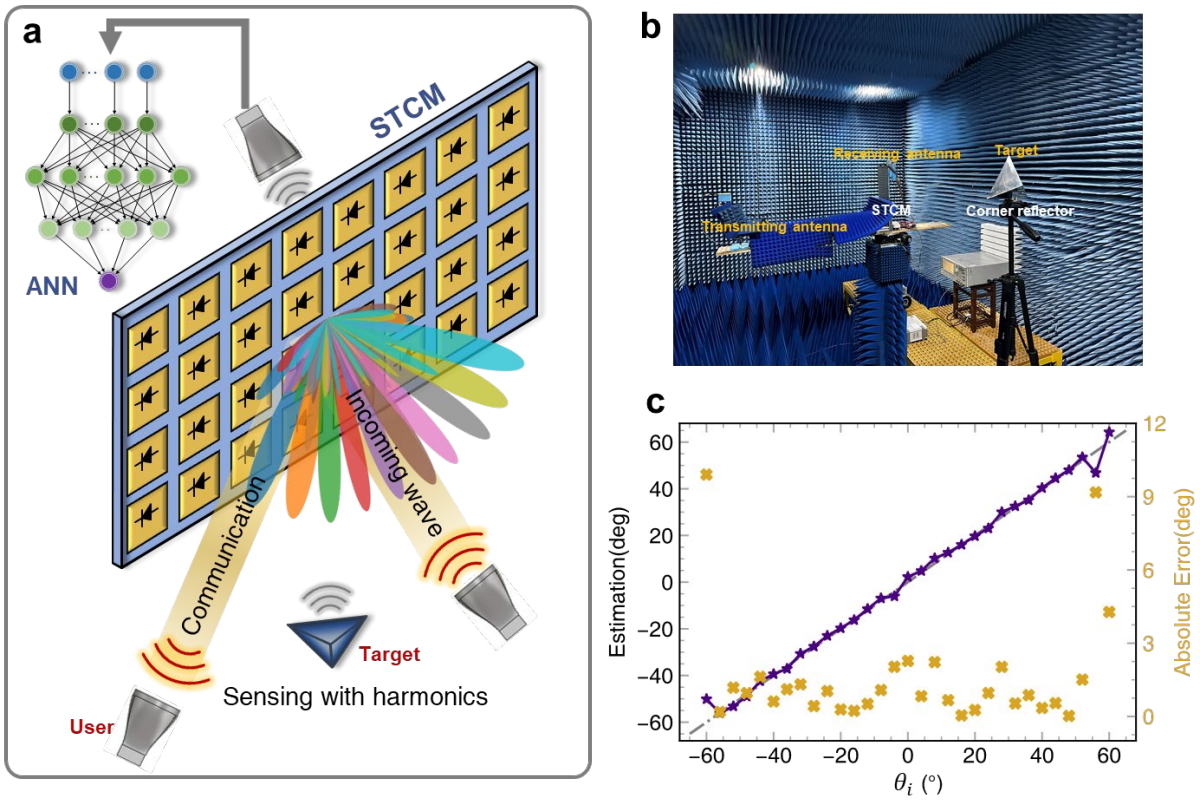


Supplementary Fig. 14 | Validation of the STCM-based ISAC system in a realistic indoor environment without absorbing materials. **a**, Experimental setup. **b**, Demodulated constellation diagrams for 10 positions of antenna ANT1 when the STCM is not operational and operational. **c**, Measured EVM for different angles of incidence when the STCM is not operational and operational.

Supplementary Note 12: Alternative STCM-Based ISAC Scheme

Supplementary Fig. 15a presents an alternative STCM-based ISAC scheme, featuring a fixed communication link and a passive target to be sensed. In this setup, the STCM modulates the incoming wave, dividing it into multiple harmonics serving as virtual nodes for sensing, while reflecting the remaining fundamental-frequency wave toward the intended communication user. The generated harmonics are reflected by the target being sensed and received by an antenna positioned on the STCM. An ANN architecture then analyzes the amplitude of the harmonics to estimate the target location. The sensing scheme is validated in an anechoic chamber as illustrated in Supplementary Fig. 15b, in which the STCM (identical to that in the main text) and the antennas are mounted on a turntable, and a corner reflector is used as the sensing target. The receiving antenna is fixed above the STCM, while the transmitting antenna is oriented perpendicular to STCM, transmitting a monochromatic plane wave at 10.3 GHz. Using the STC matrix shown in Fig. 2c, the STCM converts the incident wave into multiple harmonics. The position of the corner reflector is fixed, while the turntable is adjusted to various angles to position the corner reflector at different directions. The receiving antenna captures the

harmonics reflected by the corner reflector for spectrum analysis, and the vector of normalized amplitudes of the harmonics is input to a trained ANN architecture to estimate the location of the corner reflector. Supplementary Fig. 15c displays the estimation results and errors, demonstrating a good accuracy within the range [-50°, +50°].



Supplementary Fig. 15 | Alternative STCM-based scheme for ISAC. a, Conceptual illustration. **b**, Experimental setup. **c**, Estimation results and errors for the corner-reflector location.